Advanced Biotechnology

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CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 3

Nucleic acid sequencing (Part I)



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Lecture outline

1) Sanger sequencing
2) Next generation (Illumina) sequencing

Learning outcomes

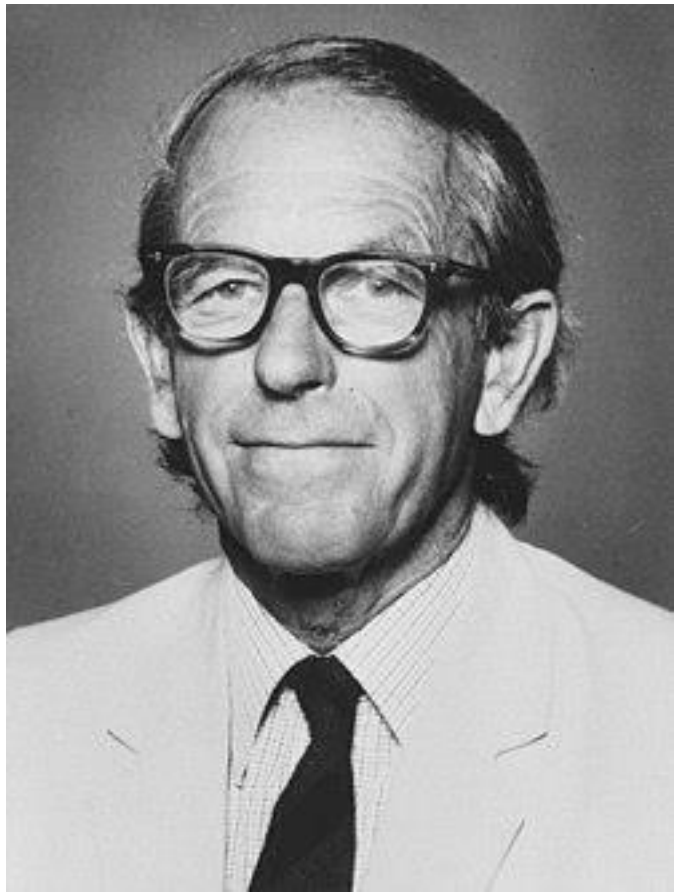
1) Describe the principles of Sanger Sequencing, including detection and separation techniques
2) Describe applicability, limitations, advantages and disadvantages of Sanger Sequencing
3) Describe steps involved in library preparation for Sanger Sequencing of genomic DNA (Shotgun sequencing)
4) Describe principles and steps involved in Next Generation Illumina Sequencing, including library preparation, Bridge PCR and Sequencing By Synthesis
5) Describe applicability, limitations, advantages and disadvantages of Illumina Sequencing

Lecture outline

1) **Sanger sequencing**
2) Next generation (Illumina) sequencing

Frederick Sanger

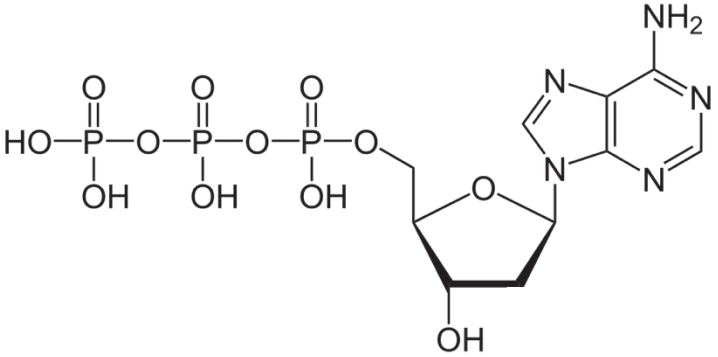
- Cambridge-educated biochemist
- Worked at Medical Research Council Laboratory of Molecular Biology
- Wond TWO Nobel Prizes in Chemistry: one for discovering the **sequence of insulin** and one for developing the first method for **DNA sequencing** (shared with Walter Gilbert and Paul Berg)
- The Wellcome Trust Sanger Institute is named after him



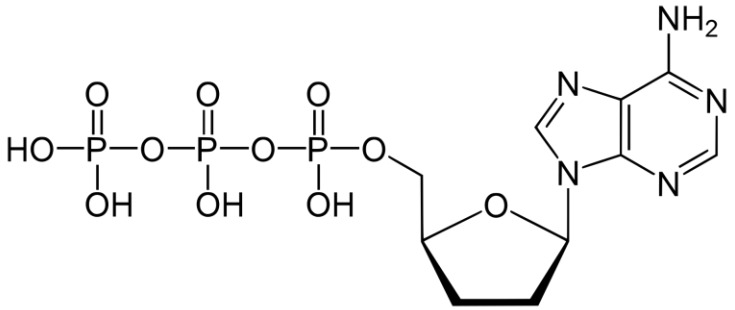
Frederick Sanger 1918-2013

Sanger sequencing

- Requires
  - A single-stranded **DNA template** (that we would like to sequence)
  - **A primer**
  - **DNA polymerase**
  - Deoxynucleotide triphosphates (**dNTPs**)
  - Dideoxinucleotide triphosphates (**ddNTPs**)
  - (other co-factors for the polymerisation reaction)

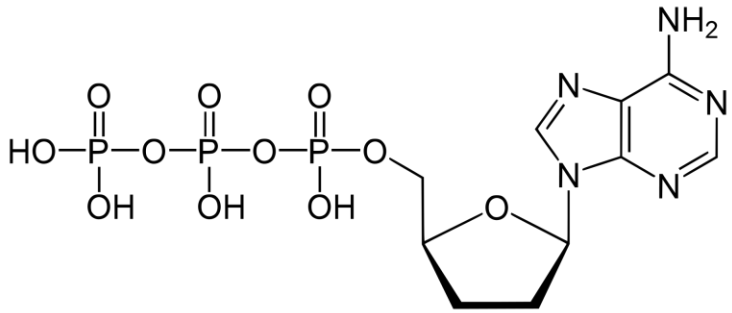


Deoxinucleotide triphosphate (dATP)



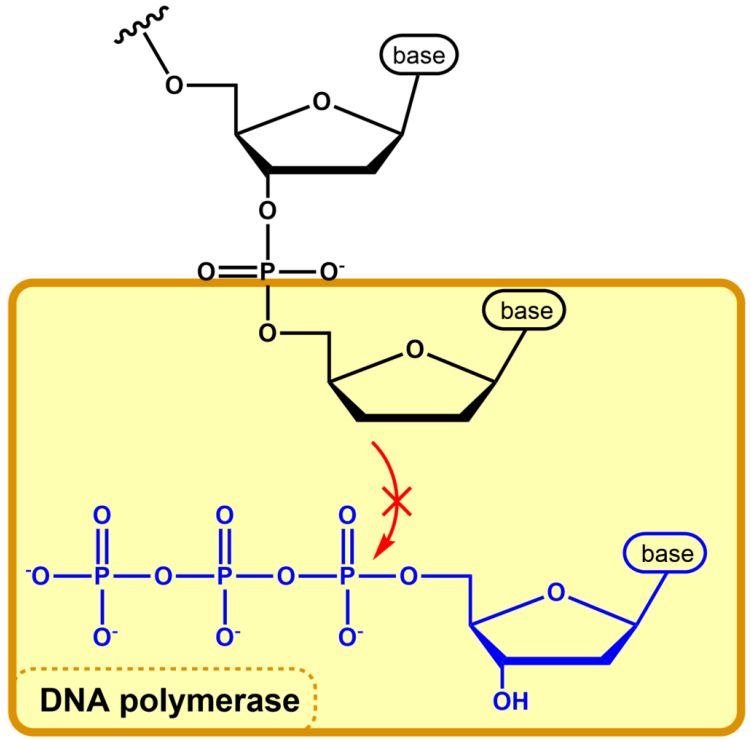
Dideoxinucleotide triphosphate (ddATP)

Sanger sequencing



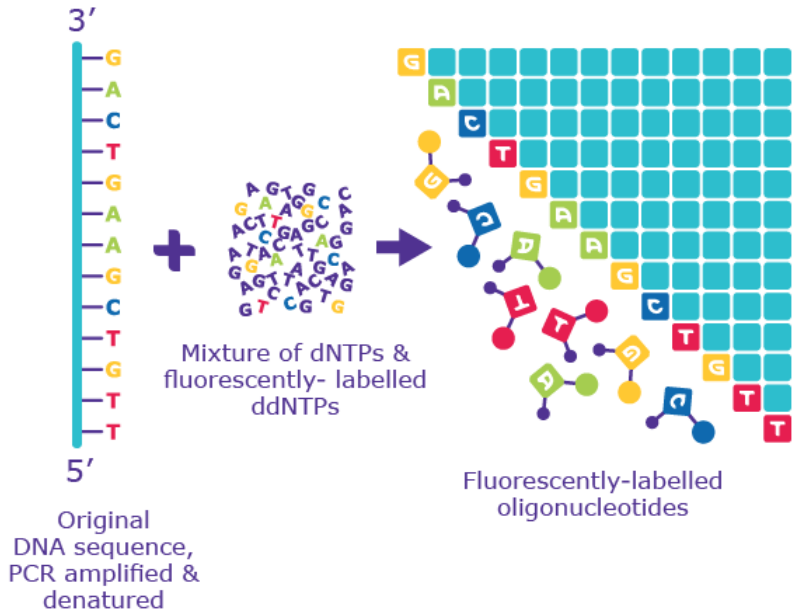
Dideoxinucleotide triphosphate (ddATP)

- ddNTPs lack the OH group on the 3' carbon of the sugar
- When a ddNTP is incorporates in the nascent DNA chain by the polymerase, elongation will terminate as no bond can be made with subsequent nucleotides



Sanger sequencing

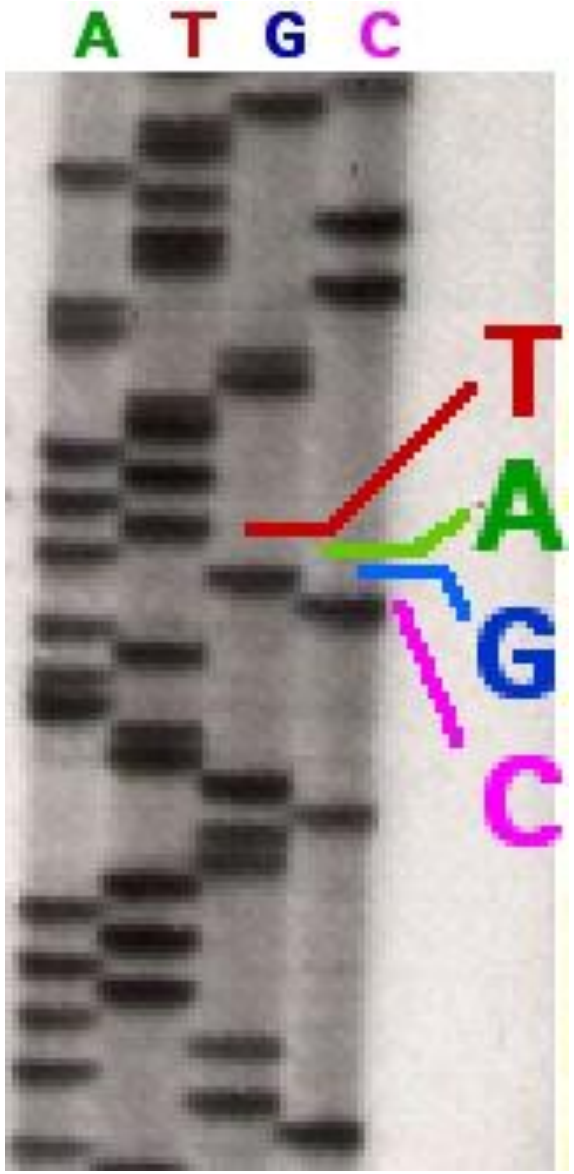
- In Sanger sequencing, a polymerisation reaction is run in the presence of a small amount of ddNTPs (~1%), alongside dNTPs
- Occasionally, a ddNTP will be incorporated in the DNA chain, which causes elongation to stop and produces a truncated product
- Electrophoresis can then be used to detect the truncated products



Credit: AAT Sigma-Aldrich

Sanger sequencing

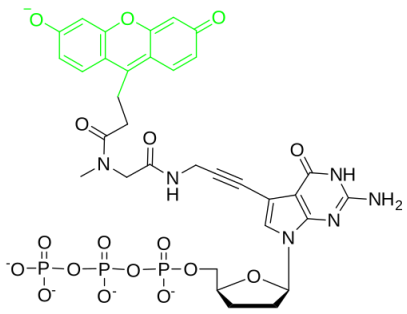
- In early implementations, one had to run **four parallel reactions**, each of which included only one of the four possible ddNTPs: ddATP, ddTTP, ddCTP and ddGTP
- In each reaction truncation would therefore occur only at the locations in the sequence matching the corresponding nucleotide
- Running electrophoresis of the four reactions in parallel enables sequence reconstruction

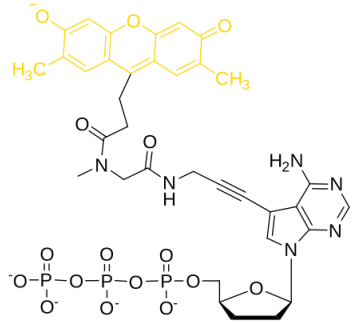


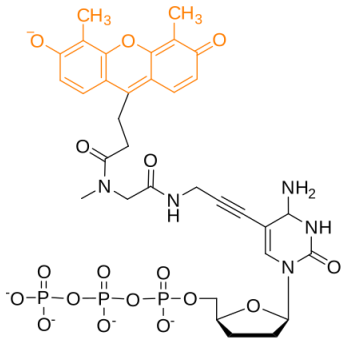
Radioactively labelled sequencing gel

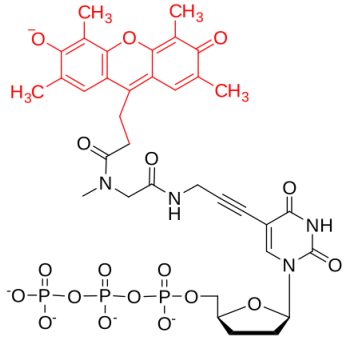
Sanger sequencing

- Introduction of fluorescently labelled ddNTPs enabled sequencing to be run as a single reaction
- The four ddNTPs are labelled with spectrally distinct fluorophores (emitting light at different wavelength)


G-505

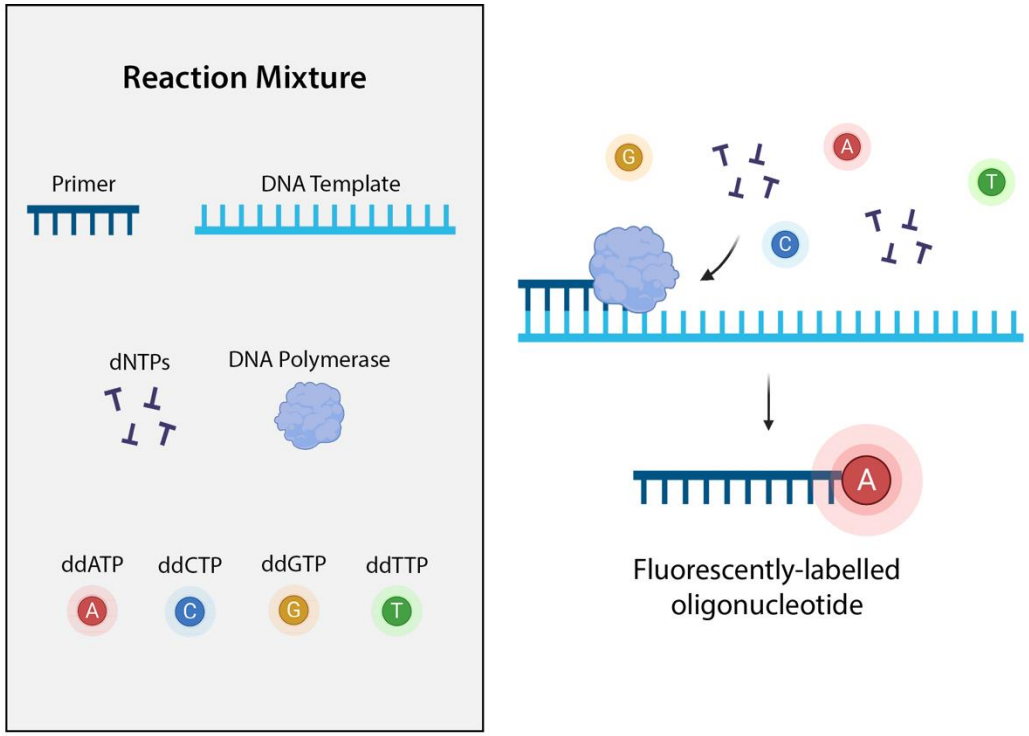

A-512


C-519


T-526

Sanger sequencing

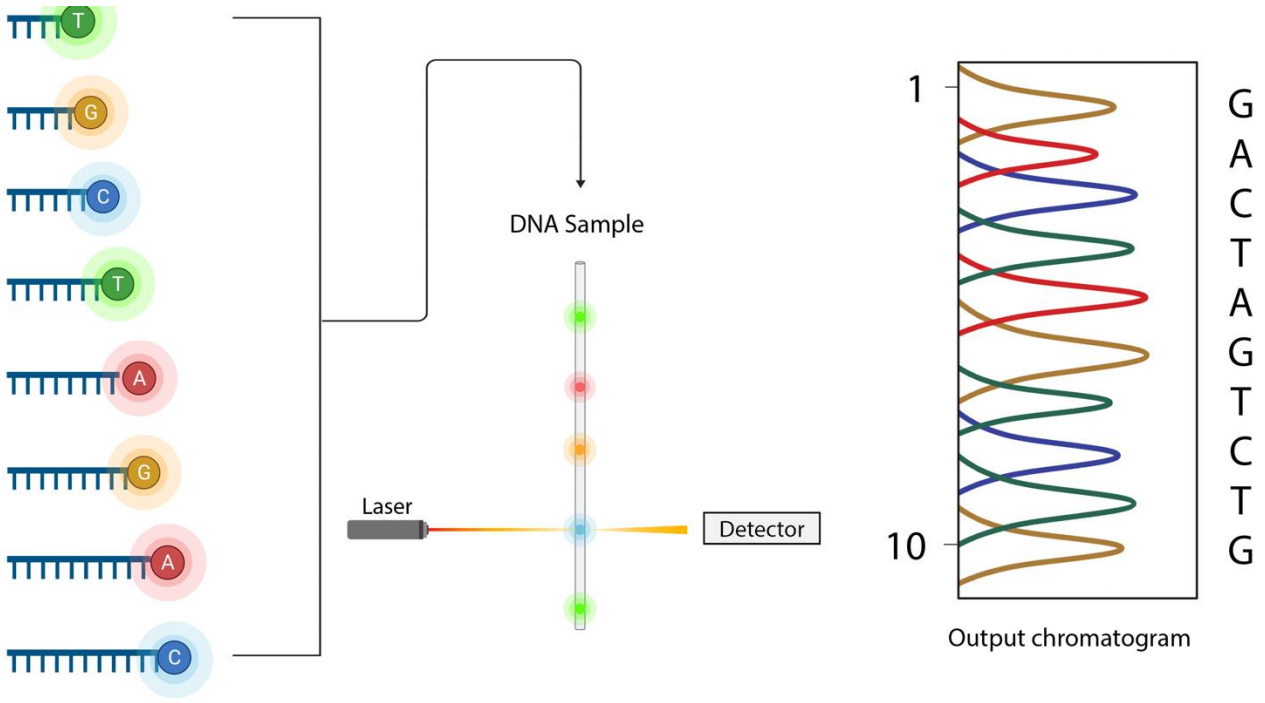
All truncated fragments will therefore be labelled with a different fluorophore depending on the final (3') base



Credit: AAT bioquest

Sanger sequencing

- Capillary electrophoresis can then be used to separate truncated products by length AND identify the final nucleotide using fluorescence
- Sequence can be read from the output chromatogram, namely from the sequence of peaks (bands) observed at the different wavelengths



Credit: AAT bioquest

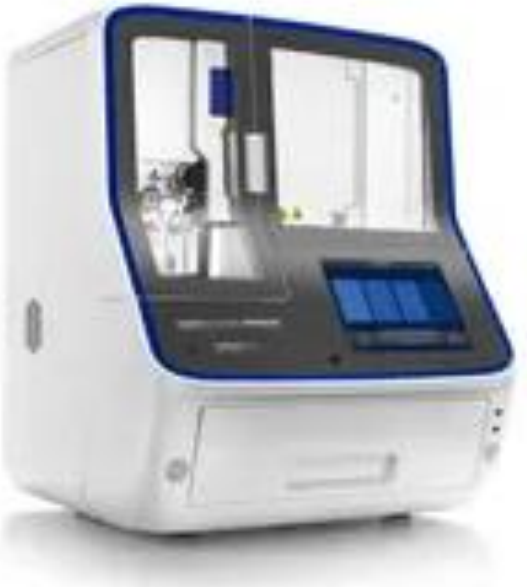
Sanger sequencing

Advantages

- The process can be automated with stand alone machines
- It is very accurate, up to 99.99%
- It can produce relatively long reads, >500 nt

Disadvantages

- Low throughput
- Cannot sequence initial bases (due to primer)
- Accuracy drops for longer constructs as it gets more difficult to distinguish products of different lengths by electrophoresis

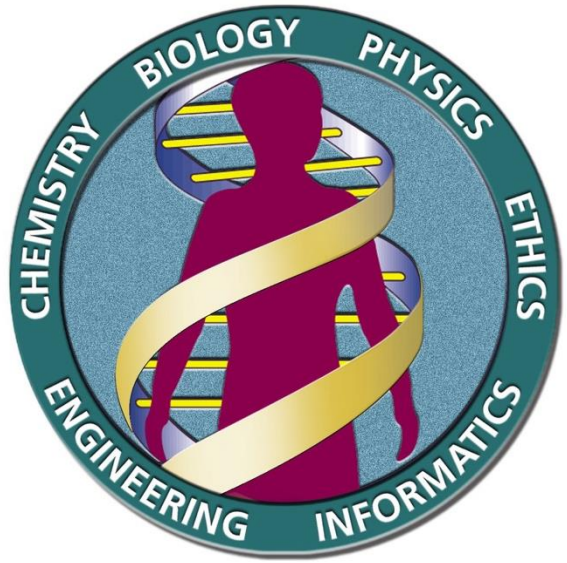


Credit: ThermoFisher

It is still used routinely when high accuracy is required and low throughput is not an issue. However, for applications that require high throughput, e.g. full genome sequencing, we now use **next generation sequencing**

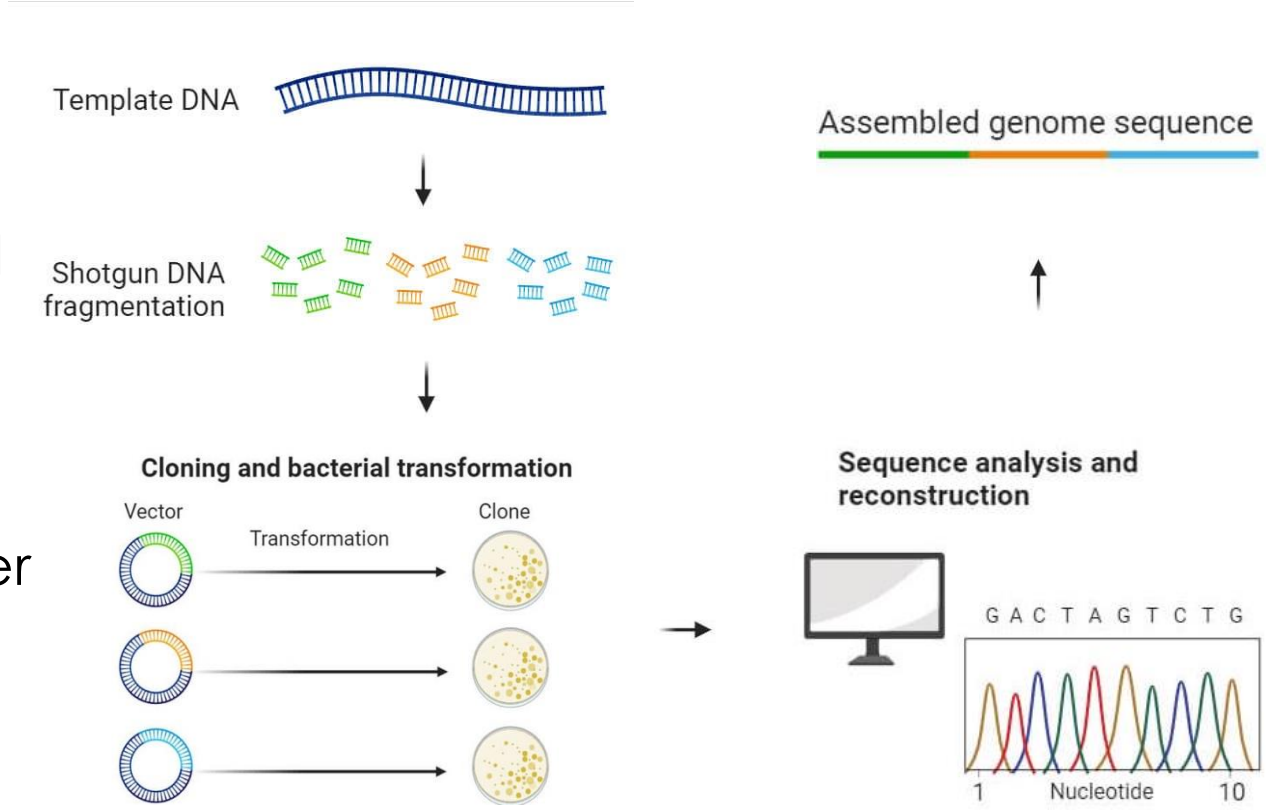
Human Genome Project

- Project carried out by multiple institutions internationally to map the entire Human Genome (3 billion nucleotides)
- Costed about $3 billion, so $1 per nucleotide
- In the UK, work done at the Sanger Institute
- Mainly funded by the US government
- Started in 1990 and completed in 2003



How to sequence genomes? Shotgun approach

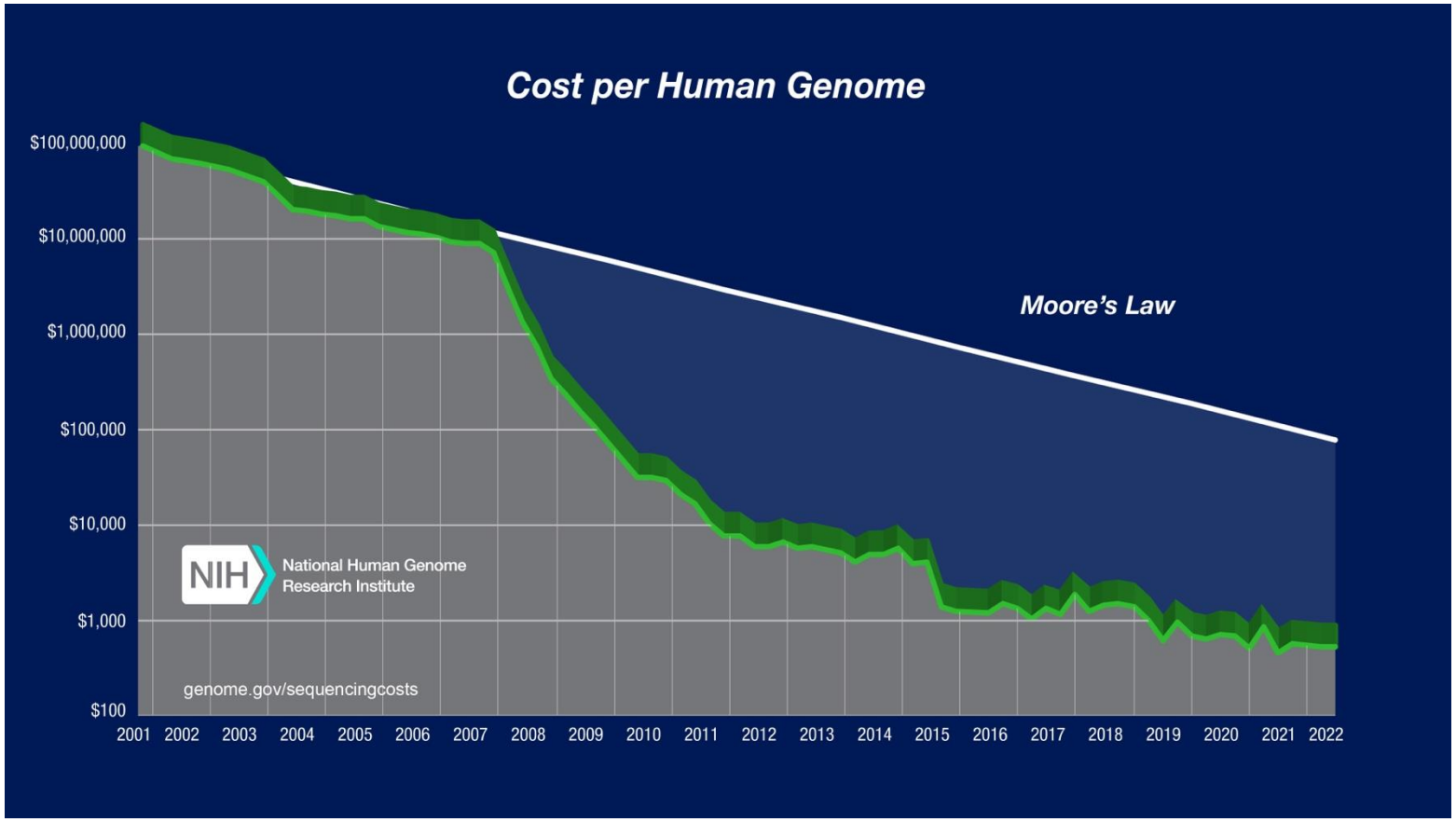
- Genomes are randomly fragmented using mechanical shearing (sonication)
- To create a **sequencing library**, individual fragments are cloned into plasmids and transformed into *E. coli* so they are amplified
- Plasmids from individual colonies are sequenced, to obtain the sequence of individual fragments
- Bioinformatics tools are used to stich together the sequences and reconstruct the entire genome
- The Human Genome Project used similar techniques



Sequencing Moore's Law

Cost of sequencing decreased exponentially (Moore's Law) until ~2007, then started decreasing at a much faster rate, plateauing at around $1000 per genome (that's ~$10^{-6} per nucleotide!)

What happened around 2007?



Lecture outline

1) Sanger sequencing
2) **Next generation (Illumina) sequencing**

Next Generation Sequencing (NGS)

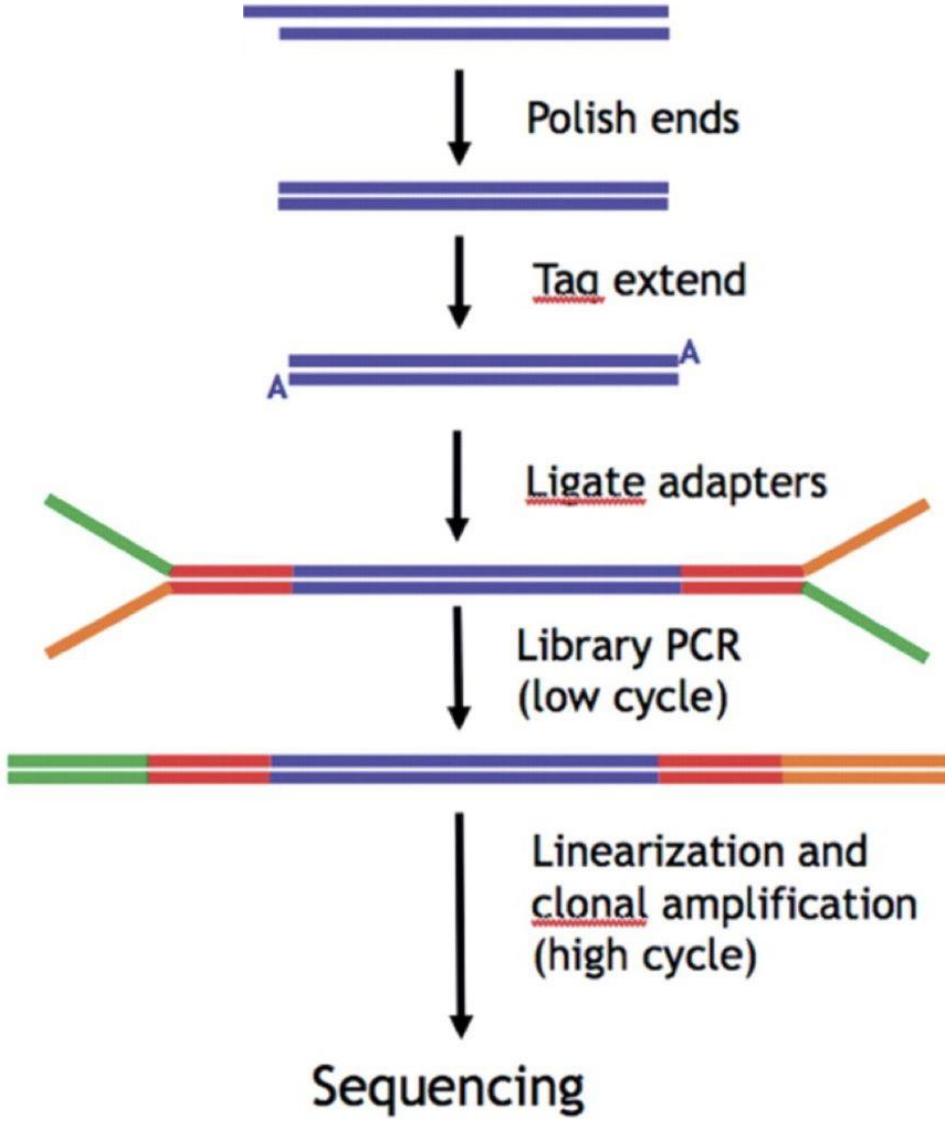
- AKA Massively Parallel Sequencing
- A collection of techniques to greatly enhance sequencing throughput
- Sanger sequencing can sample ~96 strands in one go (one well plate), NGS can produce **millions to billions of reads in one run**
- Different techniques were developed, but one emerged as the dominant approach: **Illumina Sequencing**
- Technology originated from from S. Balasubramanian and D. Klenerman in Cambridge (Chemistry), who funded a company called **Solexa**, later sold to Illumina for $600M
- Illumina has an annual revenue of 4.37 billion (2024)



Prof Sir David Klenerman (left) and Prof Sir Shankar Balasubramanian (right), Dept. of Chemistry

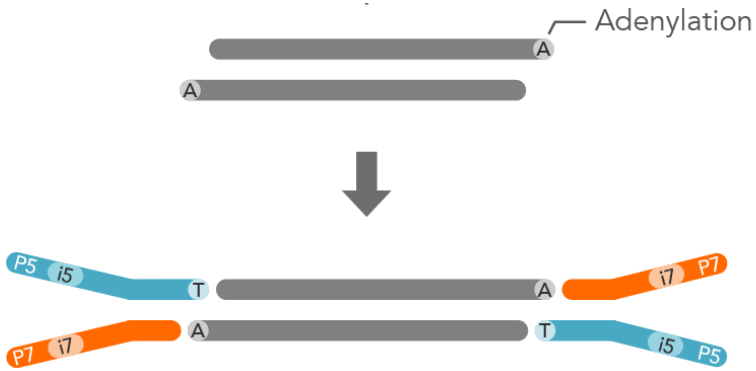
Illumina sequencing: Library Preparation

- Break the DNA to sequence in 100-200 bp fragments with sonication
- Fragments will have random short overhangs and need to be polished to produce **blunt ends**. This is done using exonucleases (to chew 5' overhangs) and polymerase (to repair 3' overhangs)



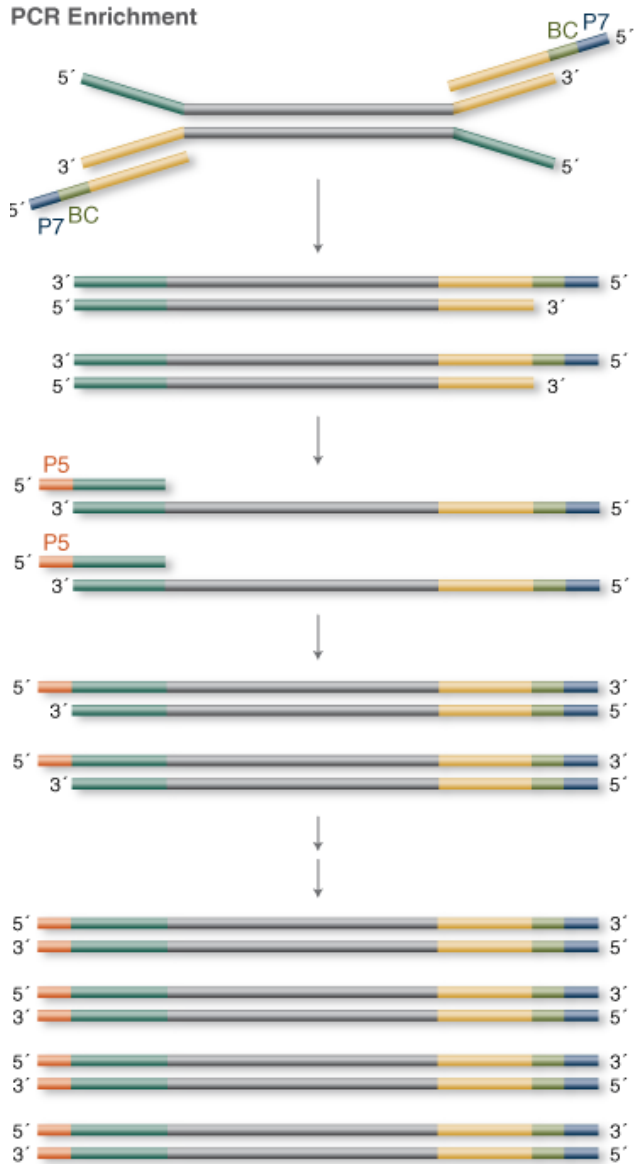
Illumina sequencing: Library Preparation

- Use Taq polymerase to add a non-templated Adenine at the 3' end. This is required to append **Y adapters**
- Y adapters have a 5' T overhang and can be **ligated** to the adenylated fragments
- Y adapters at both sides are identical. They have **identical 3' and 5' overhangs**



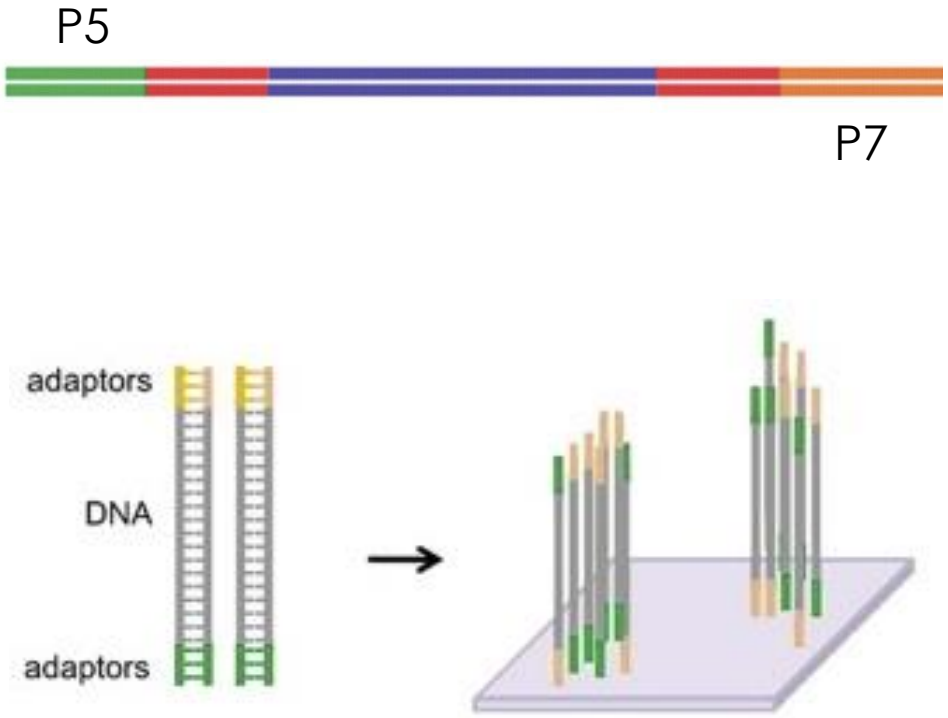
Illumina sequencing: Library Preparation

- We can use the overhangs of the Y constructs as binding sites for a forward and reverse primer in PCR
- The primers can then be used to amplify the constructs with PCR and append **different sequences at the two ends of the fragments, labelled P5 and P7**
- It is critical that all fragments have P5 and P7
- A barcode sequence (BC ~6nt) is also included



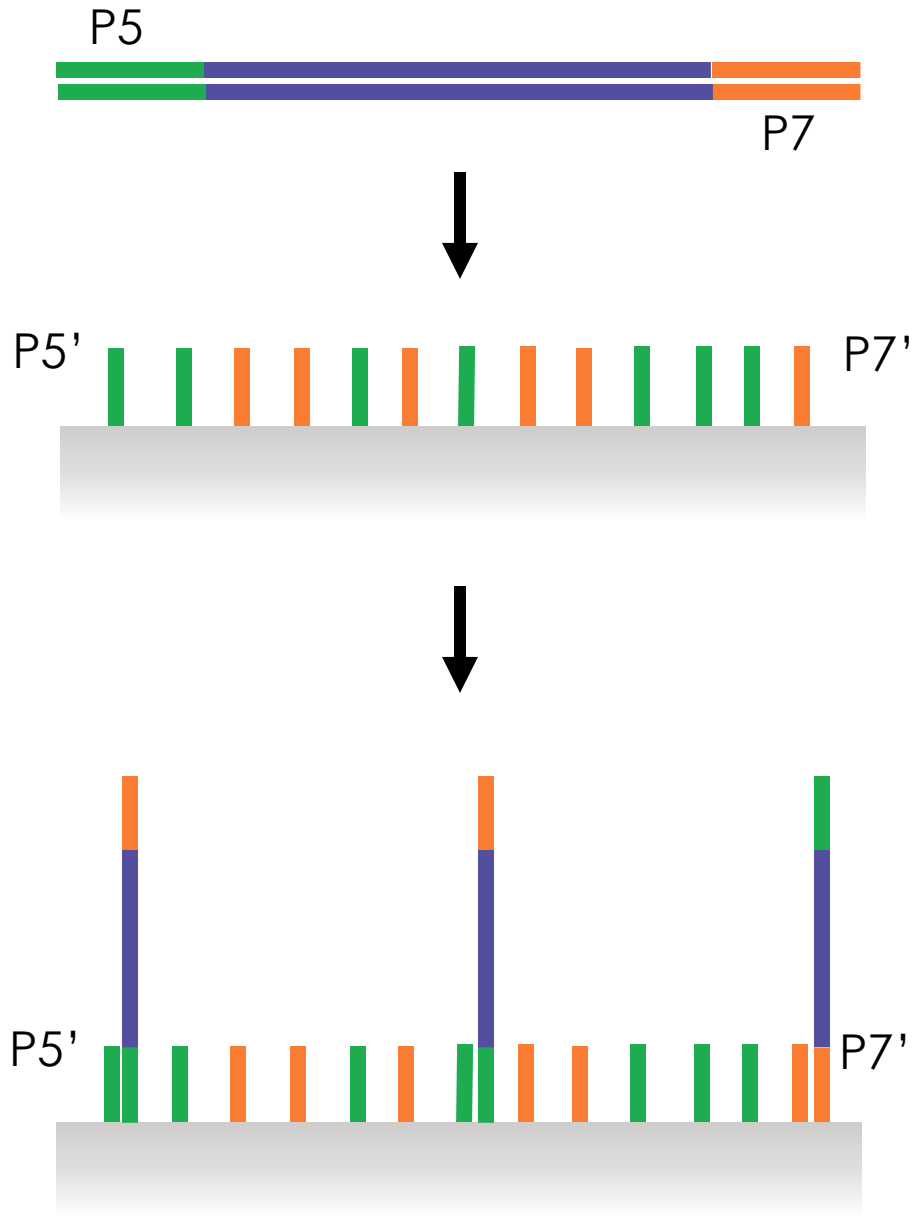
Illumina sequencing: Amplification and cluster generation

- We have now prepared the library such that each individual fragment terminates with specific P5 and P7 sequences at either end
- We would like to sequence as many as possible of these fragments **in parallel**
- To achieve this, Solexa developed a process called **Bridge or Surface PCR**, that produces millions or billions of clusters of identical fragments on a surface
- **Each cluster** (and thus each fragment) can then be sequenced through a **Sequencing By Synthesis (SBS)** approach



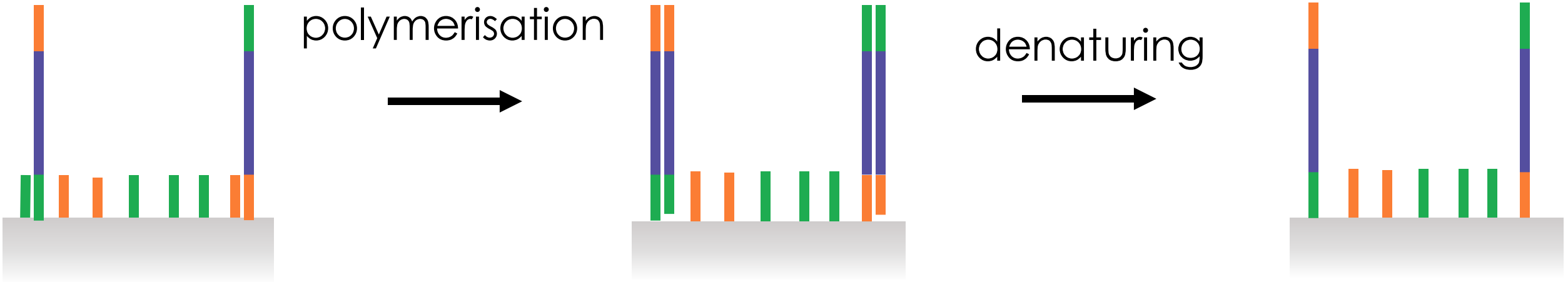
Illumina sequencing: Bridge PCR for cluster generation

- Consider a silicon-derived surface with a high density of oligos tethered, with sequences P5' and P7', complementary to the P5 and P7 regions on the fragments
- The surface (microarray) is in a **flow cell**, and can be exposed to a sequence of reagents
- First, we expose the surface to (denatured) fragments, such that they will randomly attach to the P5' and P7' oligos



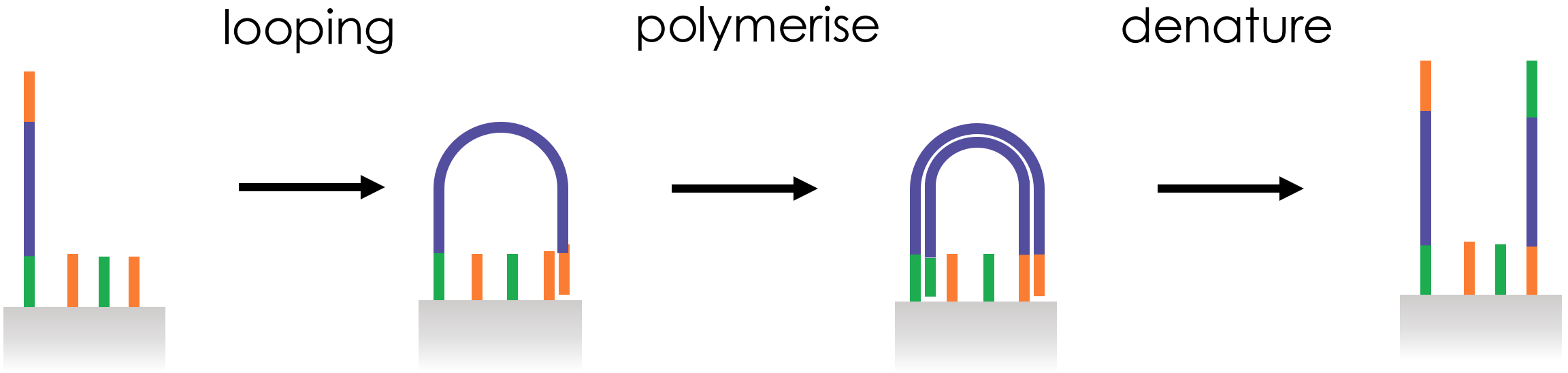
Illumina sequencing: Bridge PCR for cluster generation

- We then use a polymerase on these anchored fragments, using the P5' and P7' strands as primers
- We then de-nature the newly formed duplexes to release the initial fragments, obtaining a surface with a (relatively sparse) coverage of covalently tethered fragments



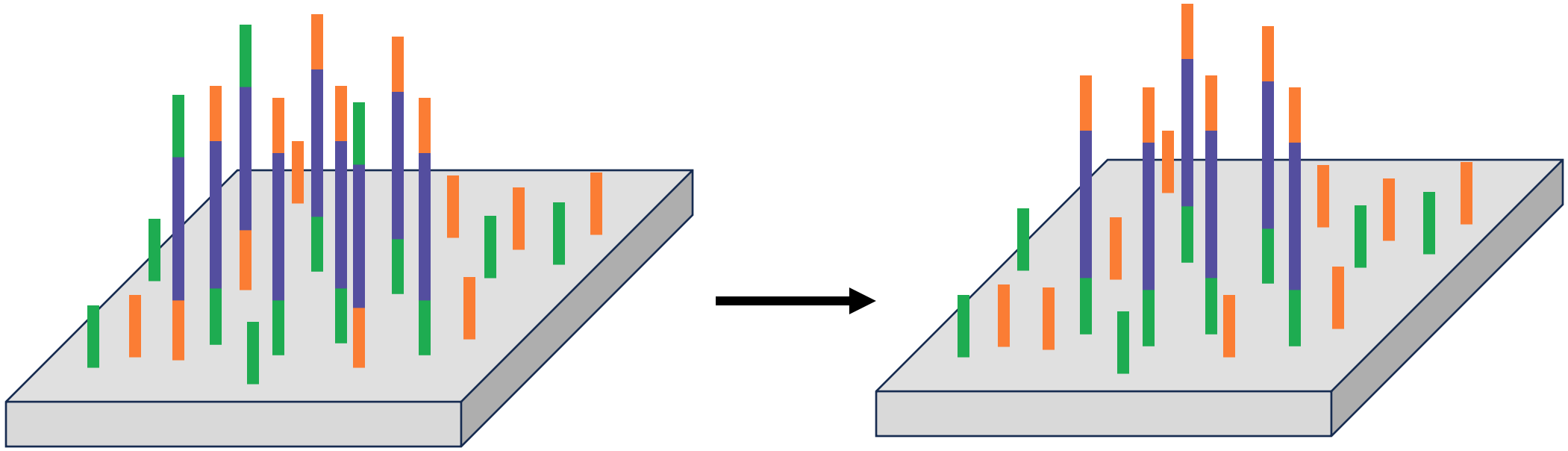
Illumina sequencing: Bridge PCR for cluster generation

- The tethered strands can then re-bind to free P7' pr P5' strands on the nearby surface, forming loops (or bridges)
- We can then polymerise and de-nature once more, which will produce a new copy of the construct



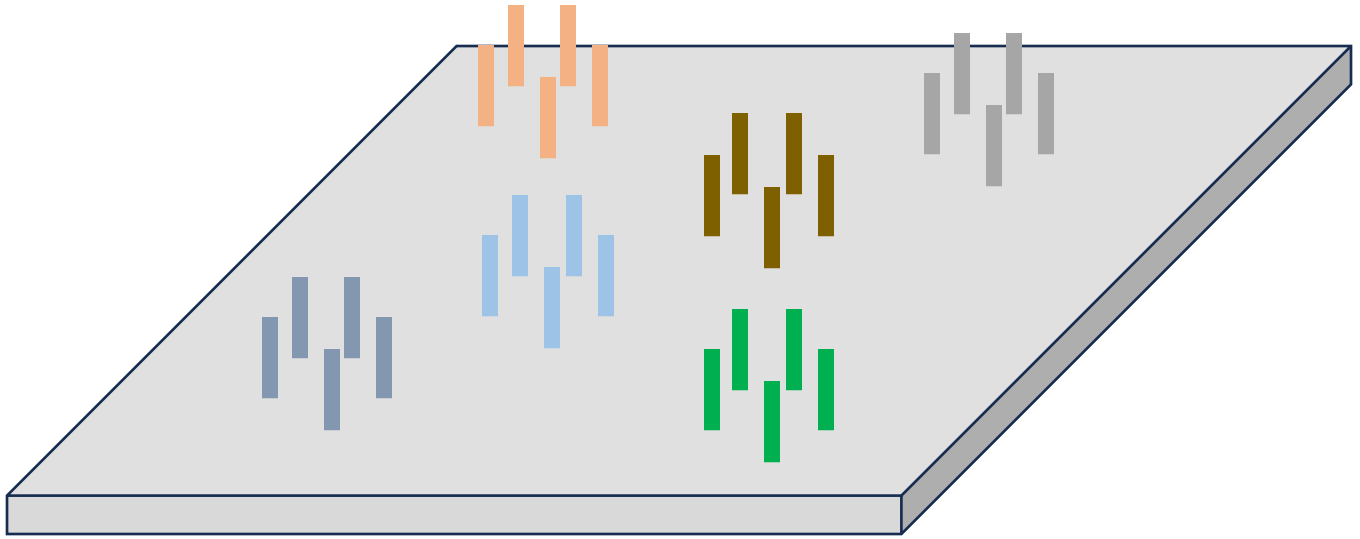
Illumina sequencing: Bridge PCR for cluster generation

- The process is repeated multiple times (25-35 times) to generate up to thousands copies in the vicinity of the original fragment, forming a cluster with **up to thousands of copies**
- In some protocols, all fragments with one orientation are (enzymatically) cleaved, leaving the cluster with fragments all with the same orientation



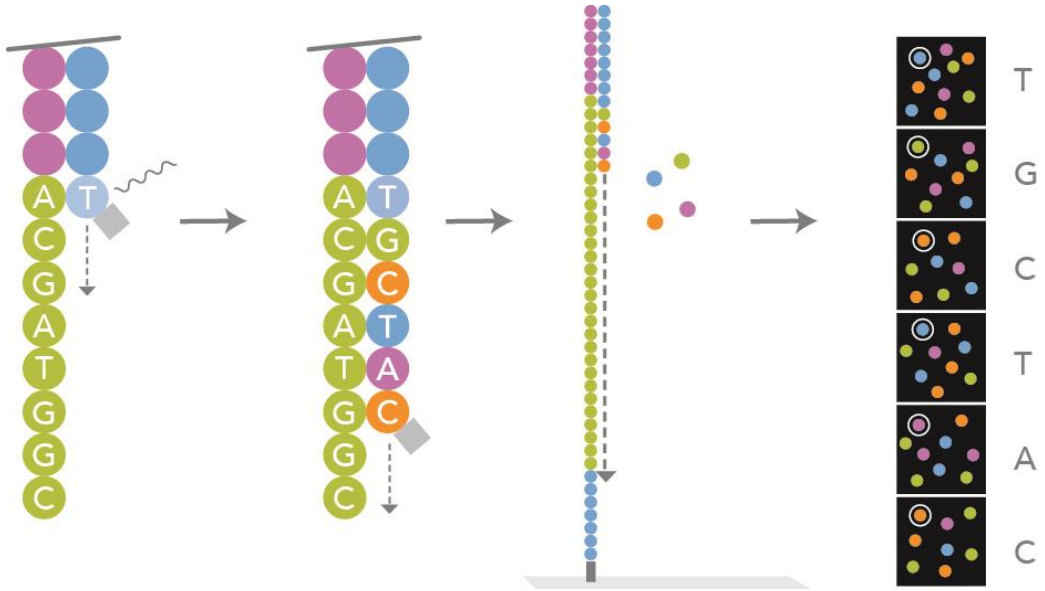
Illumina sequencing: Bridge PCR for cluster generation

- The surface will ultimately have a very large number (millions to billions) of clusters
- Each cluster contains hundreds to thousands of copies of a unique fragment, all oriented in the same way
- We can thus proceed to sequence all clusters in parallel, achieving massive throughput



Illumina sequencing: Sequencing by synthesis

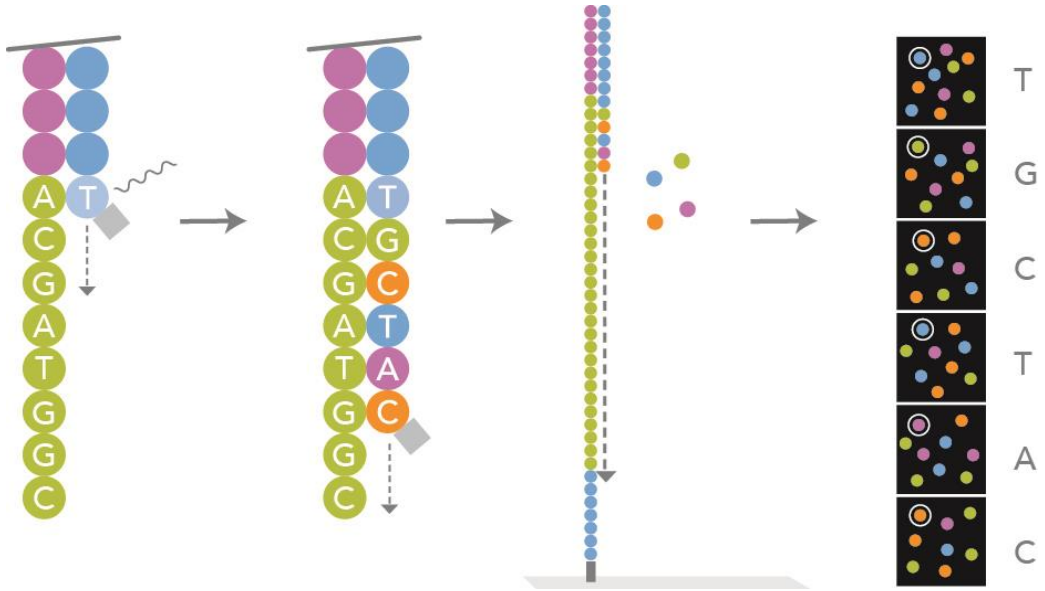
- Sequencing performed with a "sequencing by synthesis" strategy
- A primer is annealed at the 3' of the tethered fragments
- **Fluorescent versions of all four dNTPs are added at the same time**
- These are labelled with **four spectrally distinct fluorophores**



Credit: Illumina

Illumina sequencing: Sequencing by synthesis

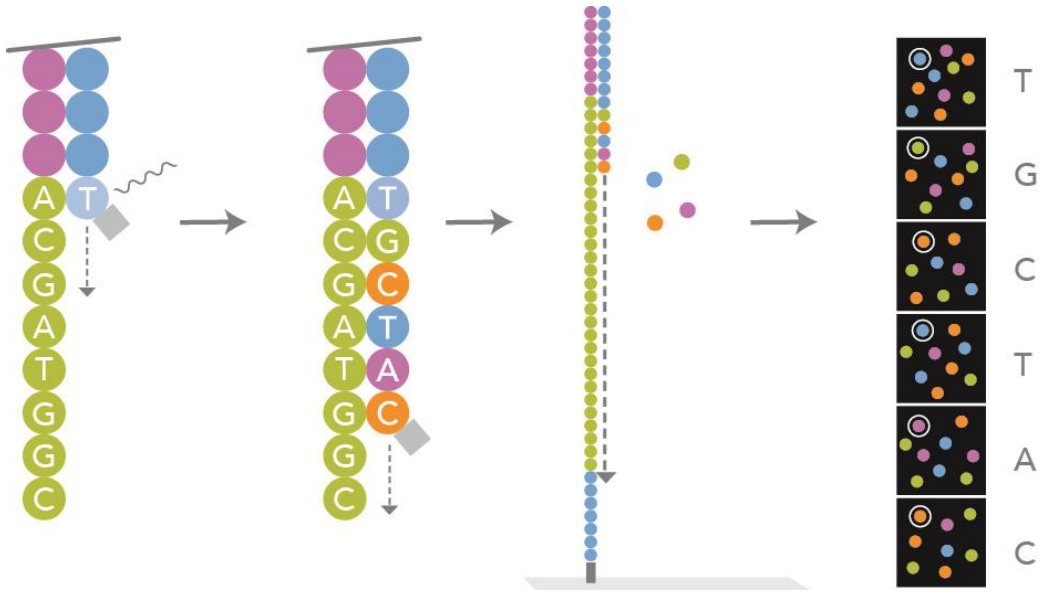
- The 3' location on the dNTPs is protected by the fluorophores, so the polymerase can only add one of the dNTPs at a time
- Therefore, at each step, only one dNTP type (dATP, dTTP, dGTP, dCTP) will be added to each individual cluster
- The entire cluster will thus be fluorescent in the colour of that one dNTP



Credit: Illumina

Illumina sequencing: Sequencing by synthesis

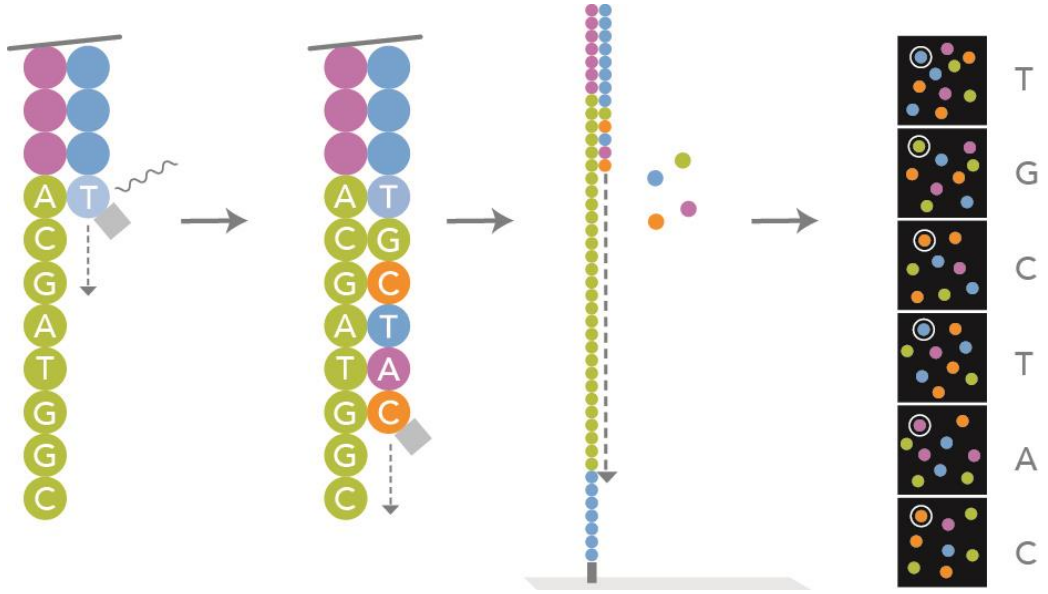
- Through fluorescence imaging, we can thus identify which nucleotide has been added to each cluster
- Imaging is carried out across the entire surface of the flow cell, to capture all clusters
- We can then de-protect the added nucleotide by removing the fluorophore and repeat the process until we have sequenced all fragments (which have similar length)



Credit: Illumina

Illumina sequencing: Sequencing by synthesis

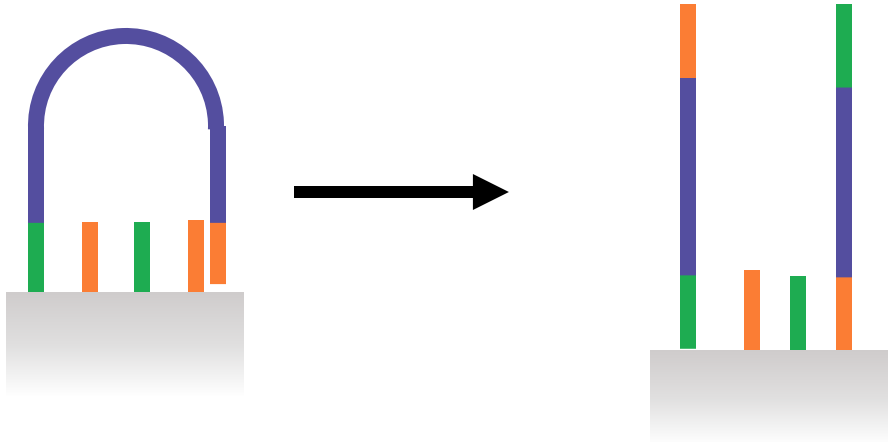
Once again: sequence information, for each cluster, is read as the sequence of "colours" emitted by the attached nucleotides

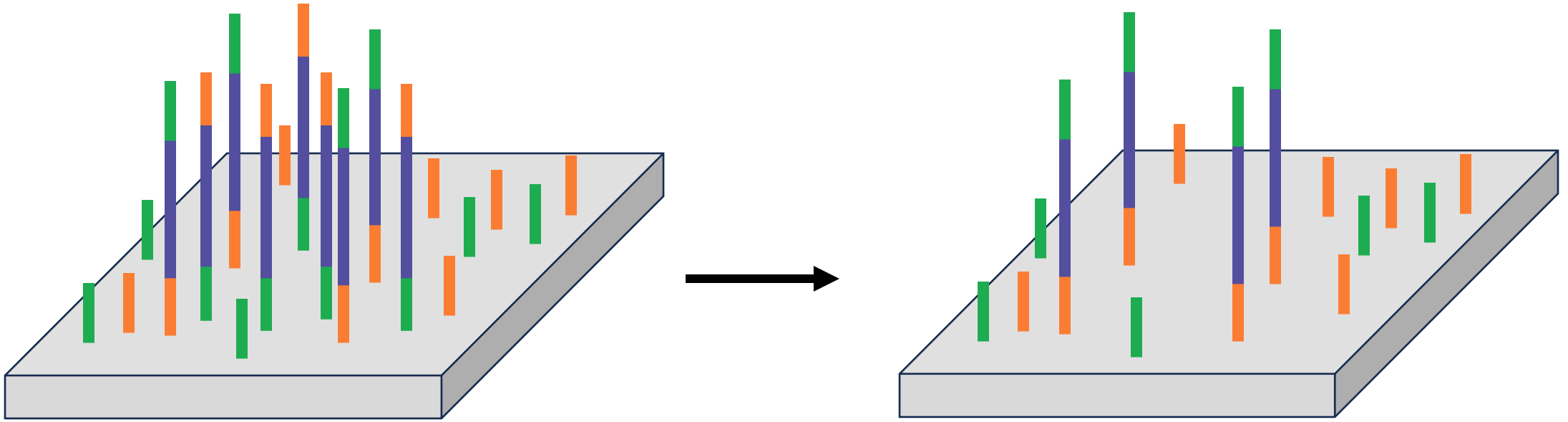


Credit: Illumina

Illumina sequencing: Sequencing by synthesis

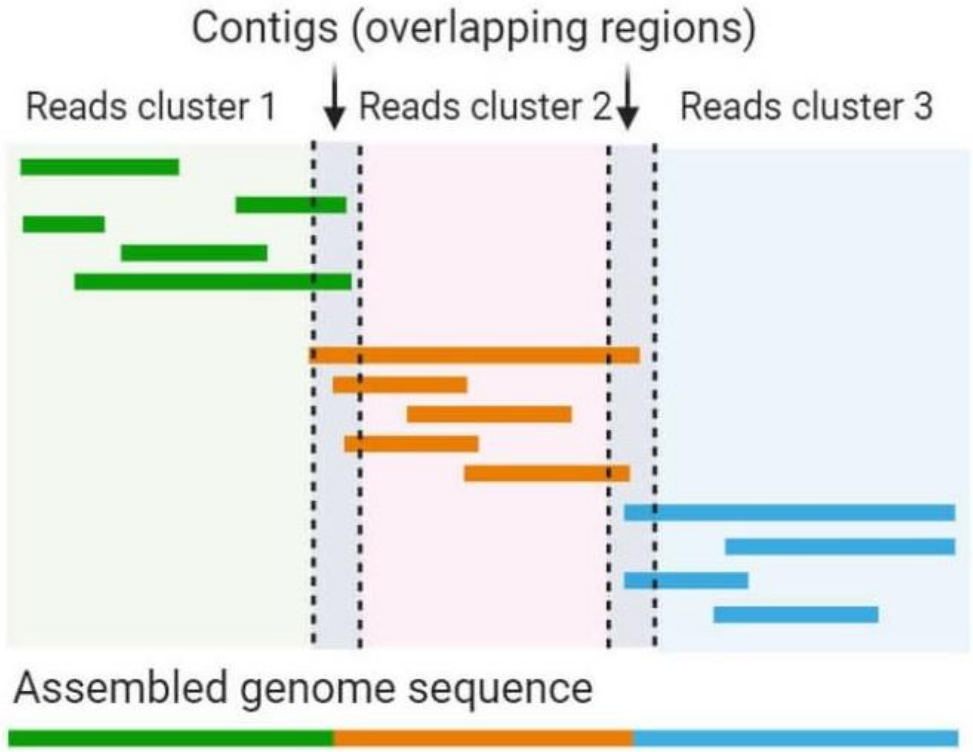
- A **second read** can be performed by repeating one round of bridge PCR to produce again the complementary fragments (with the opposite orientation) on the surface, and removing the ones we already sequenced



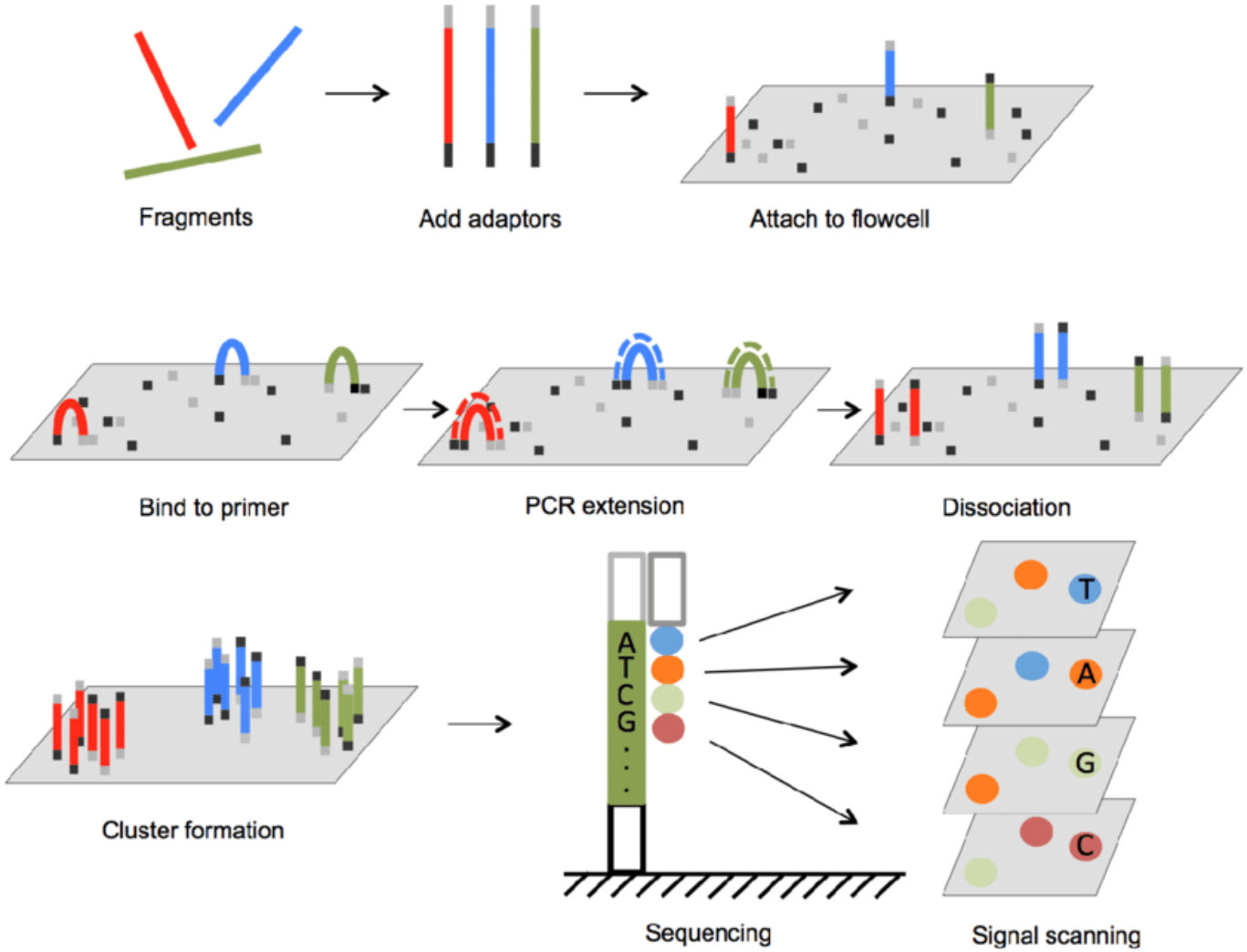


Illumina sequencing: Sequence reconstruction

- Finally, we obtain a sequence for each of the fragments on the microarray
- If these fragments came from a longer construct (a gene, a plasmid, a chromosome…) the entire sequence can be reconstructed using bioinformatics



Illumina sequencing: Overview



Illumina sequencing: Sequencing by synthesis

- All the steps can be automated because the reaction is carried out in a flow cell
- Stand alone sequencers carry out all steps



Credit: Illumina

Illumina sequencing: Pros and cons

Pros

- High accuracy, as high as 99.9% for sufficiently short fragments
- Very high throughput: billions of clusters (fragments) can be sequenced in each run
- Low sequencing cost per nucleotide

Cons

- Short reads, 50-300 nt
- Workflow is time consuming. Library preparation, bridge PCR etc need to be done. Sequencing can take multiple days
- Bridge PCR can be sequence-biased
- High cost of instrumentation (top-of-the range in the order of millions)